Output:

```
🚀 Starting Volatility Arbitrage Desk for SPY
Risk-free rate: 4.5%
Running single simulation cycle...

================================================================================
⏰ Live Update: 2026-10-11 14:54:40
📌 Underlying: SPY @ 500.00 | Expiry: 2026-10-25 (T=0.04y)
📌 Realized Vol (30d): 21.02%

🔥 TOP 5 OVERPRICED (Sell Signal) – highest IV-RV
 strike type     price       iv   spread  volume
  495.0 call 16.176101 0.349404 0.139235     127
  485.0  put  6.552842 0.349365 0.139196     201
  510.0 call  9.038055 0.345432 0.135263     253
  515.0  put 21.428974 0.344387 0.134218      50
  505.0  put 15.200862 0.343933 0.133764     342
```

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❄️ TOP 5 UNDEPRICED (Buy Signal) – lowest IV-RV

strike	type	price	iv	spread	volume
500.0	put	10.812360	0.298045	0.087876	338
485.0	call	20.805039	0.302083	0.091914	58
480.0	call	24.546725	0.304919	0.094750	370
500.0	call	11.918232	0.306096	0.095927	222
495.0	put	8.813884	0.306764	0.096595	87

📊 Total liquid contracts: 18
📊 Avg spread (bps): 1132.09 bps
📊 Std spread (bps): 183.37 bps
📊 Max overpriced: 13.92 vol pts
📊 Max underpriced: 8.79 vol pts

🔴🔴🔴 STATE: WORST (Extreme Overpricing) 🔴🔴🔴
⚡ [SIM] Speed command sent: 0.0%

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```
⚡ [SIM] Speed command sent: 0.0%
💡 [SIM] Pin 18 set to HIGH (ON)
💡 [SIM] Pin 17 set to LOW (OFF)
🚨 [SIM][ALERT:CRITICAL] Max spread 13.92% > 6%. Halting.
📈 [SIM][EXECUTION] SELL 10 contracts @ strike 495.0 (call)
💡 [SIM] Pin 25 set to HIGH (ON)
💡 [SIM] Pin 25 set to LOW (OFF)
📊 Underlying: 500.00 | RV: 21.02%
📊 Avg Spread: 1132.09 bps | Max Over: 13.92% | Min Under: 8.79%
------------------------------------------------------------------------------

🛑 Simulation complete. Shutting down desk gracefully.
⚡ [SIM] Speed command sent: 0.0%
💡 [SIM] Pin 17 set to LOW (OFF)
💡 [SIM] Pin 18 set to LOW (OFF)
💡 [SIM] Pin 25 set to LOW (OFF)
```

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